

Fixed charges of arbitrary sign: what survives and what fails

Duy T. Nguyen*

Independent Researcher, Ho Chi Minh City, Vietnam

Abstract

Fixed-charge models assume the charge is paid, not received. Wholesale electricity clears at negative prices, and a negative price turns an activation charge into a payment received. We ask what survives. The vertex argument does: for a fixed-charge integer program whose continuous constraint matrix is totally unimodular, an optimal solution is an integral vertex of the cell obtained by conditioning on its support, whatever the signs. That argument is elementary; what is not is where it becomes necessary. A fixed charge is piecewise concave in Zangwill's sense exactly when it is paid, so the classical machinery stops precisely at the sign change. The structural properties it is usually invoked to deliver do *not* survive. The vertex bound constrains only coordinates lying strictly between 1 and their upper bound, and we exhibit instances with negative charges whose unique optimum runs every activity it uses at the minimum level 1: the fractional-period property of Florian–Klein and Love, the spanning-forest bound for fixed-charge transportation and the spanning-tree bound for fixed-charge flow each fail. Algorithms resting on the classical properties therefore cannot simply be carried over. The $O(T^4)$ recursion of Akbalik, Penz and Rapine is correct under the concavity it assumes; applied outside it, we give a four-period instance on which its state space cannot represent the optimum and it loses 16.7%. Total unimodularity cannot be dropped. The same analysis exposes a modelling error in the natural mixed-integer formulation.

Keywords: fixed-charge problems, negative prices, total unimodularity, lot sizing, integral polyhedra

1. Introduction

Write the cost of using activity j at level $x \in [0, u_j]$ as

$$\varphi_j(0) = 0, \quad \varphi_j(x) = s_j + c_j x \quad (x > 0), \quad (1)$$

with a fixed charge s_j and a marginal rate c_j . In applications both are a price multiplied by a technical coefficient, $s_j = p_j \sigma_j$ and $c_j = p_j \gamma_j$ with $\sigma_j, \gamma_j > 0$, so both inherit the sign of p_j .

5 Wholesale electricity clears below zero, and increasingly so as variable renewable capacity grows [10]. There a fixed charge is received rather than paid.

The sign of s_j decides the curvature. For $x > 0$ and $0 < \lambda < 1$,

$$\varphi_j(\lambda x) - \lambda \varphi_j(x) = s_j(1 - \lambda), \quad (2)$$

*Corresponding author. ORCID: 0009-0004-0732-2093

Email address: duynguyen.das@gmail.com (Duy T. Nguyen)

so φ_j is concave on $[0, u_j]$ exactly when $s_j \geq 0$. When $s_j < 0$ it lies strictly below its own chord, which makes it *convex*; what (1) loses at the origin is not convexity but *lower semicontinuity*, since $\liminf_{x \downarrow 0} \varphi_j(x) = s_j < 0 = \varphi_j(0)$.

That distinction matters, because concavity and lower semicontinuity are exactly the hypotheses the structure theory runs on. Florian and Klein [3] assume concavity; Love [7] and Swoveland [11] assume piecewise concavity. Koca, Yaman and Aktürk [5] assume piecewise concavity together with lower semicontinuity. Akbalik, Penz and Rapine [1] invoke Love’s property explicitly under concave costs. The same model has a polyhedral literature [2], which does not use the structural property at all; and the concave case with arbitrary capacities is NP-hard [4], so the polynomial results are all for restricted capacities. Ou [8] imposes no sign restriction in his model statement, but the structural property he uses is attributed to Swoveland and is proved there for piecewise concave costs, so the hypothesis is inherited by the proof.

Two qualifications are due at once. First, [5] already anticipate a repair: their §3.4 models minimum order quantities by setting $p_t(x) = \infty$ on $(0, L) \cup (C, \infty)$, and solves that case in $O(n^6)$. With $L = 1$ the effective domain becomes $\{0\} \cup [1, C]$, which restores lower semicontinuity and leaves a piecewise concave cost whose breakpoints $\{0, 1, C\}$ are time-invariant and fixed in number — precisely the condition their polynomial algorithm requires. Negative charges are therefore not outside their framework; they are outside the framework as it is usually instantiated. The polynomial solvability of the $L = 1$ case in [5] is not at issue here. What fails, and what Sections 4–5 are about, is the transfer of the classical structural properties, and of the algorithms built on them, to charges of either sign. Second, [7] is closer than it first appears. His model already carries production bounds $x_i^L \leq x_i \leq x_i^U$, and his conclusion excepts production equal to “zero or its upper or lower bound”; with $x_i^L = 1$ that exception set is $\{0, 1, x_i^U\}$, which is exactly the conclusion of Corollary 3 below. What the corollary adds is that this holds *without* concavity — Love assumes his cost concave on $[0, \infty)$, which (1) is not when $s_j < 0$. The classical property in the usual $x_i^L = 0$ setting is a different statement, and Section 4 shows it fails.

In itself, the proof of Theorem 2 is three lines — conditioning on the support leaves an *affine* function on a bounded polyhedron whose vertices are integral by Hoffman–Kruskal — and we claim no depth for it. But the standard route to a result of this shape is genuinely closed, and Proposition 4 shows why. Zangwill [12] is the canonical reference here, and it is how [7] reaches the lot-sizing result. His theory is not confined to continuous costs: his Theorem 6 minimises any *piecewise concave* function over its dominant set, and such functions may jump. Proposition 4 shows that (1) lies in that class precisely when $s_j \geq 0$, so the sign of the charge is the exact boundary of the theory rather than an obstruction we happen to meet.

Contribution.. What is new here is negative and, we think, more useful. Theorem 2 and Corollary 3 survive arbitrary signs. Section 4 shows that the three classical structural properties they are invoked to deliver do not, with a witness in each setting whose *unique* optimum violates the classical statement. Section 5 carries the failure into an algorithm: the $O(T^4)$ recursion of [1], correct under the concavity it assumes, cannot be transferred beyond it, and loses 16.7% on a four-period instance. Proposition 6 shows total unimodularity cannot be dropped. Remark 8 records a modelling error the same fact makes visible.

2. The problem class

50 Let A be a totally unimodular matrix, b an integral right-hand side, $u \in \mathbb{Z}_{>0}^n$, and consider

$$(FC) \quad \min \Phi(x) = \sum_{j=1}^n \varphi_j(x_j) \quad \text{over } x \in \mathbb{Z}^n, \quad Ax \leq b, \quad 0 \leq x \leq u, \quad (3)$$

with φ_j as in (1) and $s_j, c_j \in \mathbb{R}$ of arbitrary sign. Write $\text{supp}(x) = \{j : x_j > 0\}$, and for $S \subseteq \{1, \dots, n\}$ define the cell

$$P(S) = \{x \in \mathbb{R}^n : Ax \leq b, \quad x_j = 0 \ (j \notin S), \quad 1 \leq x_j \leq u_j \ (j \in S)\}. \quad (4)$$

The constant 1 in (4) is where integrality enters, and it is the source of everything in Section 4: for continuous variables the set $\{x_j > 0, j \in S\}$ is not closed and carries no vertex guarantee.

55 3. The vertex theorem

Lemma 1. *Let x^* be feasible for (3) and $S = \text{supp}(x^*)$. Then $x^* \in P(S)$; every integral point of $P(S)$ is feasible for (3) with support exactly S ; Φ agrees on $P(S)$ with the affine function $\sum_{j \in S} s_j + \sum_{j \in S} c_j x_j$; and $P(S)$ is a non-empty bounded polyhedron all of whose vertices are integral.*

Proof. x^* is integral with $x_j^* \geq 1$ on S , so $x^* \in P(S)$. Conversely any $x \in P(S)$, integral or not, has $x_j \geq 1 > 0$ exactly on S , hence $\Phi(x) = \sum_{j \in S} (s_j + c_j x_j)$; the first sum is constant on $P(S)$, so this is affine there. Boundedness is immediate from $0 \leq x \leq u$. The constraint matrix of $P(S)$ is A with rows of the identity adjoined, which preserves total unimodularity, and every right-hand side — entries of b , u , 0 and 1 — is integral; by Hoffman–Kruskal the polyhedron is integral. $\square$

The affineness is the whole of the argument. The only part of Φ that is neither convex nor concave as a whole is the switch at the origin, and conditioning on the support switches it off: once we know which activities are used, the fixed charges are a constant and nothing non-linear remains. No curvature is needed, so no sign restriction is needed either.

Theorem 2. *Problem (3) has an optimal solution $\hat{x}$ that is an integral vertex of $P(\text{supp}(\hat{x}))$, for fixed charges and marginal rates of arbitrary sign.*

70 *Proof.* Let x^* be optimal and $S = \text{supp}(x^*)$. By Lemma 1, $P(S)$ is a non-empty bounded polyhedron on which Φ is affine, so Φ attains its minimum over $P(S)$ at a vertex $\hat{x}$, and $\Phi(\hat{x}) \leq \Phi(x^*)$. Again by Lemma 1, $\hat{x}$ is integral and feasible with support exactly S , so $\Phi(\hat{x}) \geq \text{OPT} = \Phi(x^*)$. Hence $\hat{x}$ is optimal and is a vertex of $P(S) = P(\text{supp}(\hat{x}))$. $\square$

75 **Corollary 3.** *Let $\hat{x}$ be as in Theorem 2, let A_t be the rows of A tight at $\hat{x}$, and let $Q = \{j : 1 < \hat{x}_j < u_j\}$. Then the columns of A_t indexed by Q are linearly independent; in particular $|Q| \leq \text{rank } A_t$.*

Proof. Suppose $d \neq 0$ is supported on Q with $A_t d = 0$. For small $\varepsilon > 0$ the points $\hat{x} \pm \varepsilon d$ satisfy the tight rows with equality and the slack ones by continuity; the bounds hold because every coordinate of Q is strictly interior; and coordinates outside S are untouched, since $Q \subseteq S$. So $\hat{x}$ is the midpoint of a segment in $P(S)$, contradicting that it is a vertex. $\square$

80 **Proposition 4.** *Let $X = [0, u_j]$ and let φ_j be as in (1). Then φ_j is piecewise concave in the sense of [12] if and only if $s_j \geq 0$. His Theorem 6 therefore applies to a fixed charge that is paid and to no fixed charge that is received.*

Proof. Zangwill calls F *continuous piecewise concave* (CPC) on X when $F = \max_i g_i$ for finitely many continuous concave g_i , with *basic sets* $B_i = \{z \in X : F(z) = g_i(z)\}$ and *terminal set* 85 $T = \bigcup_{a \neq b} (B_a \cap B_b)$; and F is *piecewise concave* (PC) when it is the pointwise limit of CPC functions $\{F_k\}$ that share one terminal set T and satisfy $F_k = F_1$ on $E[X] \cup T$.

Suppose $\{F_k\}$ is such a sequence for φ_j . Each F_k is a maximum of finitely many continuous functions, hence continuous; and $F_k = F_1$ on $E[X] \cup T$ together with $F_k \rightarrow \varphi_j$ pointwise gives $F_1 = \varphi_j$ there.

90 Since $0 \in E[X]$ we get $F_1(0) = \varphi_j(0) = 0$. As F_1 is continuous and $s_j + c_j x \rightarrow s_j \neq 0$ as $x \downarrow 0$, choose $\delta \in (0, u_j)$ with $|F_1| < |s_j|/2$ and $|s_j + c_j x| > |s_j|/2$ on $[0, \delta]$. Any $x \in T \cap (0, \delta)$ would satisfy $F_1(x) = \varphi_j(x) = s_j + c_j x$, violating both bounds; hence $T \cap (0, \delta) = \emptyset$.

Fix k . The basic sets of F_k are closed and cover X , and no two of them meet in $(0, \delta)$, which is connected; so exactly one basic set meets $(0, \delta)$ and F_k agrees there with a single continuous concave 95 g^k . By continuity, $g^k(0) = F_k(0) = F_1(0) = 0$. Concavity with $g^k(0) = 0$ gives, for $x \in (0, \delta)$ and $\lambda \in (0, 1)$,

$$g^k(\lambda x) \geq \lambda g^k(x) + (1 - \lambda)g^k(0) = \lambda g^k(x).$$

Letting $k \rightarrow \infty$ and using $g^k \rightarrow \varphi_j$ pointwise on $(0, \delta)$,

$$s_j + c_j \lambda x \geq \lambda(s_j + c_j x), \quad \text{that is} \quad s_j(1 - \lambda) \geq 0,$$

so $s_j \geq 0$, contradicting $s_j < 0$.

Conversely let $s_j \geq 0$, put $K_0 = s_j/u_j + |c_j| + 1$ and

$$F_k(x) = \min\{(k + K_0)x, s_j + c_j x\}, \quad k \geq 1.$$

100 Each F_k is a minimum of two affine functions, hence continuous and concave, hence CPC as a family with the single member F_k itself. That family has no second member, so no two basic sets meet and the terminal set is $\emptyset$ for every k ; condition (2) then only asks for agreement on $E[X] = \{0, u_j\}$. There $F_k(0) = \min\{0, s_j\} = 0 = \varphi_j(0)$, using $s_j \geq 0$, and $(k + K_0)u_j > s_j + c_j u_j$ for every $k \geq 1$ by the choice of K_0 , so $F_k(u_j) = s_j + c_j u_j$ independently of k . Finally, for fixed $x \in (0, u_j]$ we have 105 $(k + K_0)x > s_j + c_j x$ once k is large, so $F_k(x) \rightarrow s_j + c_j x$, while $F_k(0) = 0$; thus $F_k \rightarrow \varphi_j$ pointwise and $\{F_k\}$ is a defining sequence. $\square$

110 The sign of the fixed charge is therefore not merely a technical obstacle but the exact boundary of the class Zangwill's theory covers. That is why the classical route is available for the received wisdom and closed here, and it is the precise sense in which a charge that is received is a different object rather than the same object with a hypothesis relaxed.

4. What does not survive

Corollary 3 constrains only the coordinates with $1 < \hat{x}_j < u_j$. The classical structural properties constrain those with $0 < \hat{x}_j < u_j$. Under non-negative charges the difference is immaterial in practice, because a positive fixed charge penalises opening an activity and optima consolidate. 115 Under negative charges it is the whole story: a negative fixed charge *rewards* opening, so optima

open everything they can and drive volumes to the minimum. Each witness below has a *unique* optimum, so no choice of optimal solution repairs the classical statement, and each respects the coupling $s_j = p_j \sigma_j$, $c_j = p_j \gamma_j$ with $\sigma_j, \gamma_j > 0$ of Section 1 — the failures are not an artefact of letting the fixed charge and the marginal rate carry opposite signs.

120 *Lot sizing.* Let x_t be production in period t , $u_t = C$, and let the rows of A be the prefix sums bounded below by cumulative demand and above by cumulative demand plus a storage bound; A is an interval matrix, hence totally unimodular. Take $T = 3$, $C = 3$, zero demand, storage bound 2, and prices $p = (-1, -1, 1)$ with $\sigma_t = \gamma_t = 1$, so $s_t = c_t = p_t$. The unique optimum is $x = (1, 1, 0)$, with prefix sums $(1, 2, 2)$ and value -4 . Rows 2 and 3 are tight and row 1 is not, so
125 no inventory point separates periods 1 and 2, both of which produce strictly between 0 and C : the fractional-period property of [3, 7] fails on the unique optimum. Here $Q = \emptyset$, so Corollary 3 holds vacuously.

Fixed-charge transportation. Let A be the 2×2 bipartite incidence matrix with supplies and demands $(2, 2)$, $u_{ij} = 2$ and $s_{ij} = c_{ij} = -1$. The unique optimum, of value -8 , sends one unit
130 on each of the four arcs. No arc is strictly interior, so again Corollary 3 is vacuous, while the four positive arcs exceed the classical bound $m + n - 1 = 3$ and form a cycle.

Fixed-charge min-cost flow. The same instance read as a flow problem on four nodes with $b = (2, 2, -2, -2)$ gives four positive arcs against the spanning-tree bound $n - 1 = 3$.

So the sign hypothesis cannot be removed from the classical statements; it can only be removed
135 from the vertex statement, which is weaker. Section 5 shows the difference is not academic.

5. An algorithm that breaks

Corollary 3 is sometimes read as a licence: an algorithm whose correctness uses non-negativity only through a vertex argument should keep working when the charges go negative. The licence is void whenever the algorithm uses the *classical* property rather than the vertex property, and that
140 is the common case.

The recursion of [1] is $O(T^4)$, is correct under the concave costs it assumes, and builds its state space directly on Love's property — which it states, after [7], in terms of periods that are *fractional* in the classical sense $0 < x_t < P$: within a subplan it assumes exactly $K = D_{u,v-1}/P$ periods produce at full capacity P and one fractional period absorbs the remainder. Take $T = 4$, $C = 5$,
145 storage bound 4, demand $(0, 0, 0, 4)$ and $s_t = c_t = -1$. Exhaustive enumeration gives optimal value -12 , attained by a plan using level 1 in several periods. The best plan the subplan decomposition can represent has value -10 : an error of 16.7%. The state space cannot express the optimum, so no amount of care in the recursion recovers it.

Remark 5. A second instance in the same family, $T = 5$, $C = 4$, reports a relative error of 564.5%.
150 We do not quote it as the headline: its optimal value is -0.31 , so the ratio is inflated by a near-zero denominator, and its *absolute* loss of 1.75 is smaller than the 2.00 above. Relative errors on objectives that change sign are not informative.

6. Total unimodularity cannot be dropped

Proposition 6. *There is an instance of (3) without total unimodularity whose unique optimum is
155 not a vertex of its cell, so the conclusion of Theorem 2 fails.*

Proof. Take the odd-cycle system $x_1 + x_2 \leq 6$, $x_2 + x_3 \leq 7$, $x_1 + x_3 \leq 6$, whose constraint matrix has determinant 2, with $u = (5, 5, 5)$, $s = (1.32, 2.51, 0.55)$ and $c = (-2.02, -2.06, -1.19)$. Exhaustive enumeration over the 113 feasible integer points gives the unique optimum $\hat{x} = (2, 4, 3)$ of value -11.47 , the runner-up being $(3, 3, 3)$ at -11.43 . Its support is $\{1, 2, 3\}$, so its cell is $\{x : Ax \leq b, 1 \leq x \leq 5\}$; the first two rows are tight there and no bound is, so the tight system has rank $2 < 3$ and $\hat{x}$ is not a vertex. $\square$

This shows some hypothesis is needed, not that total unimodularity is the weakest one; whether balanced or perfect matrices suffice is open.

Remark 7. The weaker odd-cycle instance one first reaches for — $x_i + x_j \leq 3$ with $1 \leq x \leq 3$ and objective $-\sum_j x_j$ — does *not* serve. Its cell has continuous optimum -4.5 at $(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$ while the best integral point is -4 at $(1, 1, 2)$, so the affine-minimum step reaches a fractional vertex; but $(1, 1, 2)$ is itself a vertex of the cell, and the conclusion survives. That example breaks the argument, not the theorem.

Remark 8 (A modelling consequence). The natural mixed-integer formulation of (1) writes $x_j \leq u_j y_j$ with y_j binary. For $s_j \geq 0$ that suffices, because setting $y_j = 1$ with $x_j = 0$ only adds cost. For $s_j < 0$ it does not: the solver sets $y_j = 1$ and $x_j = 0$ and banks the charge. Over integer x the constraint $y_j \leq x_j$ is required. On a four-period instance with prices $(1, -5, -5, -5)$, two units to be served by the horizon, $u = 2$ and unit coefficients, the guarded formulation returns -20 and the unguarded one -25 : the unguarded model reports a cost 25% below the truth, and its solution “uses” idle activities purely to collect the charge. The object it optimises is the lower semicontinuous hull of φ_j , which is the function [5] assume and (1) is not.

The unguarded pattern is the standard one. [1] write $0 \leq x_t \leq P y_t$ with no lower link, and [6] write $x_t \leq M y_t$ having just *defined* “ $y_t = 1$ if $x_t > 0$, and 0 otherwise” — a two-sided definition implemented by a one-sided constraint. Both are correct in their own settings, where the setup cost is non-negative and $y_t = 1$ with $x_t = 0$ is never optimal; we report no error in either.

Two qualifications keep this in proportion. First, the repair is not new where the issue is already live: in energy models the minimum-load constraint plays exactly this role, as in $p_t \geq P^{\min} z_t^{\text{on}} + P^{\text{sb}} z_t^{\text{sb}}$ of [9], which forbids an activated unit from consuming nothing. What is missing from the lot-sizing formulations above is a constraint the energy literature writes as a matter of course. Second, the general shape — an indicator pinned on one side only, correct because of the sign of its coefficient — is widespread and not confined to (1). The same paper [9], whose numerical study is explicitly about nonpositive prices, defines its start-up binary by $z_t^{\text{su}} \geq z_{t-1}^{\text{off}} + z_t^{\text{on}} + z_t^{\text{sb}} - 1$ with no upper pin, which is correct only because the start-up cost is a positive constant rather than a price. Our point is about (1); the pattern is broader.

7. Computational verification

Every number below is produced by a named script released with this note, runnable with no arguments. Two of them corrected claims in earlier drafts of this work.

The theorem.. Over 399 lot-sizing instances with $T \in \{4, \dots, 6\}$, $C \in \{4, \dots, 8\}$ and charges drawn from $[-3, 3]$, enumerating *every* optimal plan: no violation of Corollary 3, the bound biting on 219 of them (`redo_sec7.py`). Over 1,085 instances whose matrices were *checked* to be totally unimodular, with mixed $\leq / \geq / =$ rows and $u \in \{1, 2, 3\}$: no violation of Theorem 2 (`check_thm1_tu.py`).

How often the classical property fails.. Over 3,000 random lot-sizing instances with prices of arbitrary sign and the coupling of Section 1 imposed ($s_t = p_t\sigma$, $c_t = p_t\gamma$, $\sigma, \gamma > 0$), the fractional-period property fails on *every* optimum in 215 of them, or 7.2%; releasing the coupling raises this only to 8.6%, so the coupling is not what protects the property (`sign_coupling.py`). Separately, Corollary 3 is vacuous ($Q = \emptyset$ at every optimum) on 11.5% of mixed-sign instances against 9.2% of non-negative ones (`check_vacuity.py`). The failures are a minority, but a substantial one, and an algorithm must be correct on all instances.

Beyond the stated cost class.. The argument extends to per-period costs that are piecewise linear by conditioning on which piece each coordinate falls in as well as on the support. One point deserves care. If the coefficients of adjacent pieces are unrelated the cost *jumps* at each breakpoint, so the pieces are half-open exactly as $(0, u_j]$ is, and integrality closes them: the piece containing a used coordinate is $[\beta_k + 1, \beta_{k+1}]$, and $[1, \beta_1]$ for the first. Those are rows of the identity with integral right-hand side, so total unimodularity survives. Taking the pieces as $[\beta_k, \beta_{k+1}]$ instead does *not* work, and produced violations until the bounds were corrected. With two interior breakpoints, $T \in \{4, 5\}$, $C \in \{6, 7, 8\}$ and all four coefficient families drawn independently: 0/200 violations with non-negative coefficients, 0/200 with arbitrary signs, 0/200 with arbitrary signs and no storage bound, the check biting on 158, 172 and 191 of them respectively (`check_piecewise.py`). Independently, 266 piece-conditioned cells were solved as linear programs and none has a fractional optimum (`L1_L2.py`).

The steps of the proof.. That Φ is affine on $P(S)$ was verified on 1,269 same-support pairs, worst error 3.6×10^{-15} ; as a control, without which the argument would be vacuous, allowing the support to vary breaks affinity on 217 of 220 instances (`check_affine.py`). The condition of Corollary 3 was compared against the rank of the full active constraint system on all 6,088 feasible plans of 120 lot-sizing instances, agreeing in every case (`check_vertex_rank.py`).

8. Concluding remarks

The vertex statement survives arbitrary signs by an elementary argument for which we claim no depth — though Proposition 4 shows the classical route to it is closed, and that the sign of the charge is exactly where Zangwill’s class ends. What we think matters is that the statements it is usually invoked to deliver do not survive, that this is visible in all three classical settings, and that a published polynomial recursion — correct under its own hypotheses — cannot be transferred beyond them, losing optimality on an instance of four periods. The practical reading is the reverse of the one a vertex argument invites: a negative fixed charge does not merely relax a hypothesis, it changes which plans are optimal, from few-and-large to many-and-minimal.

Three directions remain. Whether the class extends beyond totally unimodular continuous parts is open. Proposition 4 characterises the summand φ_j , which is the object the classical hypotheses are stated about, and its necessity half lifts to Φ with s_j replaced by the charge that *switches on*. Let z be a vertex of X and d a feasible direction there, and put

$$J(z, d) = \sum \{ s_j : z_j = 0, d_j > 0 \}.$$

Then $\Phi(z + td) \rightarrow \Phi(z) + J(z, d)$ as $t \downarrow 0$, and running the proof of Proposition 4 along the ray — z is pinned because $z \in E[X]$, the terminal set avoids a neighbourhood of it, and one concave piece is active on the rest — gives $J(z, d)(1 - \lambda) \geq 0$. So Φ fails to be piecewise concave as soon

as some $J(z, d)$ is negative, the one-dimensional statement being the case $z = 0$, $d = e_j$, $J = s_j$. Both witnesses of Section 4 fail this way, the transportation one at the vertex $(0, 2, 2, 0)$ along $(2, -2, -2, 2)$, where x_{11} and x_{22} switch on together and $J = -2$. Whether $J(z, d) \geq 0$ everywhere is also *sufficient* we have not determined. And of the formulations we checked, the two lot-sizing ones [1, 6] omit the guard of Remark 8 — harmless where they stand, not where they would be reused — while the energy model [9] carries its continuous analogue as a minimum-load constraint. A search for a published formulation that both prices a fixed charge at a signed quantity and omits the guard returned nothing, so the exposure we describe is at present a property of how these models would be combined rather than one we can exhibit in print.

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